

Raised beds and bunk-bed design review

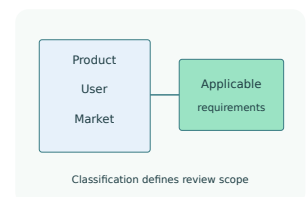
Elevated and bunk beds need specialist verification of falls, entrapment, support and access. Prepare a traceable review record; the figures are hazard maps, not construction drawings.

What you will accomplish

- Identify the product, user and market
- Review guards, mattress and openings as a system
- Require traceable verification evidence

Identify the product, user and market

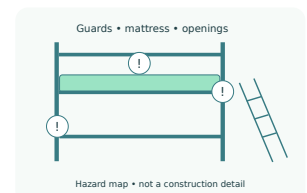
Identify the product definition, intended user and market. US reference 16 CFR 1513 addresses elevated children's beds within its scope and requires the warning against upper-bunk use below age 6. Reaching 6 does not itself establish suitability. Verify current requirements for the intended market and product; the absence of a lower bed does not automatically exclude a loft bed from review.



Classify first; a US reference does not replace local applicability review.

Review guards, mattress and openings as a system

Mark guards relative to the specified mattress, end openings, wall interfaces, ladder route and supports. A thicker mattress changes the exposed guard height; do not improvise an added rail. Entrapment assessment depends on opening shape, location and defined test methods, not one general gap rule. Never use a child's head or body as a test probe.



Review the mattress, guards and openings together; a sketch is not a test result.

Require traceable verification evidence

Link applicable requirement, drawing revision, product/mattress identity, verifier, result and outstanding action. CPSC guidance explains accepted third-party testing, children's product certification and additional US requirements; establish what applies in the project's market. Obtain assembly, labeling and service instructions. A visual prototype or one adult standing on a bed is not a substitute for required system testing.

Requirement	Product	Evidence
Scope	Model	Test
Reference	Revision	Result

An image is not test evidence

Requirement + identified configuration + traceable result.

Worked case and decision

The exercise proposes an upper bunk for a 4-year-old and provides only an unidentified image. Reject that allocation and review an appropriate non-elevated sleeping alternative with product verification. Even changing the user's age would not resolve the separate absence of system evidence.

Decision: reject the upper-bunk allocation and obtain evidence for a suitable alternative.

Check understanding

Do an image and one informal load trial establish safety?

- No; traceable system verification is needed
- Yes, if it looks strong
- Yes, if the child is older than 6

Correct: No; traceable system verification is needed. Classification, guards, mattress, openings, supports and applicable tests concern the identified system.

Lesson questions

Is a loft bed without a lower bunk automatically excluded?

No. Check the applicable product definition and scope rather than relying on its marketing name.

Practice

Prepare an elevated-bed review register covering classification, hazards, applicable requirements, missing evidence and disposition, without a construction detail.

Product, user and market

Hazards, requirements and evidence

Missing information and disposition

Review your work against the criteria. Saving it does not constitute technical project approval.

- Market applicability explicit
- No invented gaps/capacities
- No acceptance without evidence